Alice Clara Augustine

Agriculture and Pharma Biotechnologist, Semantic Technologist, HealthCare Informatician
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Key Competency Areas

- Biotech Product Pipeline experience 23 yrs.
- Ontology modelling (knowledge capture)
- Design Thinking
- Knowledge Graph Engineering

Education

- Ph.D.: Bio-Sciences (1999).
- Masters: Botany: Microbiology specialization
- Master's in Health Informatics (2021)
- Product Management UCLA (2023)

Certification & Achievements

- 8 Principal author Peer reviewed publications
- 6 co-authored peer-reviewed publications
- 10 Patents
- MIT certified "Design Thinking"
- CIO award at Amgen 2017
- Innovation Award nominee 2018
- Innovation Award IPTC 2022

Technical Skills

Laboratory related

- Plant tissue culture related
- Techniques in Genetic transformation of plants
- Microphotography, Micro-techniques and staining methods

Computational

Bioinformatics

- Bioinformatics tools such as GCG tools (BLAST, FASTA, Tree building, Motif search, Pileup, sequence alignment etc.)
- Systems biology, Pathway mining tools

Data Science

- Data management,Information modeling
- TEXT MINING, NLP tools and document classification methods.
- Ontology/taxonomy development and tools Protege, TopBraid, Semaphore; PoolParty, STARDOG, Cambridge Semantics, CENtree, RDF/OWL, SPARQL, D2RQ, R2RML
- Knowledge Graph Engineer
- Graph based platforms (Neo4j, Triple Stores, MarkLogic
- Python (Beginner), R (Beginner) and SAS

Visualization

- Tableau and Spotfire

Operational tools

- Microsoft Office Suite
- Relational Databases & Oracle 8i, SQL, SPARQL
- Visual Basic V 6.0

Project management related

- Scrum/Agile methodology
- MS project
- RACI methodologies; LEAN
- Networking and collaboration
- Communication (written and spoken)

Professional Summary

- Alice Augustine is a Senior professional with 23 years of industry experience spanning the areas of life-science and data mining and analysis.
- Ontology evangelist for 19 years, worked extensively in areas of Agriculture and Pharmaceutical industries implementing the ontology strategy for Bayer, Amgen, and GSK.
- Worked in India and the US on global teams at Monsanto Co, Amgen Inc and currently at GSK.
- Leader with broad experiences in Gene discovery, Biotech Product advancement, decision analysis, and biotech production agriculture. Worked with extensive data (genomic, field trial, clinical trial, literature)
- Experience in project/product management, working collaboratively across functional teams to achieve business objectives, prioritization, leading by influence and technical documentation.

GlaxoSmithKline (Sept 16 2020 – Current date)

Title: Director, AI Data Ops Biological Ontologist

This role will be played in the intersection of AI/ML platforms, Data Engineering, Pharma R&D discovery, working with world-renowned researchers and data scientists, and inventing the biomedical data store of the future. In this role I:

- Lead the data modeling and semantic knowledge graph ontology creation efforts for R&D, helping in modeling data domains such as multi-omics including genomics, transcriptomics, proteomics, metabolomics, microbiomics, medicinal experiment metadata capture, high throughput screening and assay data, and scientific literature to enable the enterprise data fabric/semantic layer to make data connected and findable.
- Expertise in terminology management methods and tools such as CENtree (SciBite) ; Semaphore, PoolParty, TopQuadrant, Protege
- Demonstrated knowledge of semantic web standards (RDF, SKOS, OWL, OBO, SPARQL, SHAQL) as well as linked data principles
- •Experience with existing biomedical ontologies and data standards (ontologies in BIOPORTAL) e.g. HPO, Disease, Bioassay, MeSH, CDISC, UMLS MedDRA, SNOMED CT, UBERON, MONDO, Gene Ontology, DrugBank, Who Drug, IDMP etc)
- Ability to query data management systems of different kinds: RDBMS, NoSQL, Graph databases and triple stores (e.g., Marklogic and StarDog)
- Knowledge Graph: Create the plan and method for the Data Engineering team to access, source, cleanse, merge, ETL, and map disparate data to the ontologies from 1,000s of data sources both internal and external to GSK.
- Demonstrate internal and industry presence: speak at conferences, contact senior leaders in vendor companies, and collaborate with GSK partners and collaborators.
- Assessment of new/innovative ideas and development of prototypes for high throughput data classification (structured and unstructured).
- Evangelizing the use of ontology models for capture of knowledge for the purposes of data integration, AI (ML & NLP), NLG, searching and tagging the universe of data.
- Accountable for creating relevant data principles, standards, and best practices in alignment with W3C, and communicating them to business partners and IT leadership.

Amgen Inc (Feb 22, 2016 – August 21 2020)

Title: Principal Information Architect – Reference Data Platform Product Owner
Roles as Ontology Lead

- Developing the vision, overarching guidelines, and frameworks for organizing, representing, and accessing knowledge (content and expertise) including development of overall knowledge/content strategies and work-plans. Knowledge Graph technology assessment and implementation.
- Set-up business process for information modeling, taxonomy, ontology engineering, data quality, data flows and knowledge graph population.
- Assessment of new/innovative ideas and development
- Evangelizing the use of taxonomy/ontology models for capture of knowledge for the purposes of data integration, machine learning, NLP, NLG searching and tagging the universe of data at Amgen
- Accountable for creating relevant data principles, standards, and best practices in alignment with W3C, and communicating them to business partners and IT leadership
- Integrations across-products, and third-party content linking, onboarding adopting relevant external data models and standards, content meta-data, text and data mining, semantic content enrichment, concept definition, business keywords, etc.).
- Service Ownership - Owning vocabulary management software and service delivery (Smartlogic Semaphore)
- Being accountable for ensuring that the Ontology program leverages leading practices, agreed standards, and aligns with internal and public Ontology programs

Projects at Amgen

Enterprise Taxonomy Strategy and Content (2016-2020)

- Defining the semantics (taxonomy and ontologies) strategy, priorities and vision for digital content classification, creation and maintenance of taxonomies, ontologies.
- Socializing the Enterprise Reference data strategy- Modular and Linked approach of relating data using reference data
- Provide enterprise/central ownership across various functional areas on key enterprise ontology assets Product, Location, Study and Party.
- Build and implement strategy for delivering this enterprise metadata to internal and external users through vetted consumption patterns (using APIs, SPARQL and data delivery methods).
- Creation of Amgen Standard Product Ontology, to service >75 downstream applications and to enable auto-classification of content systems.
- Creating Managing and Maintaining the Public and Internal taxonomies that are related to Product, Location, Study and Party

Key contribution: Was responsible for enterprise adoption of consistent ontology/knowledge graph management standards, processes, architecture and supporting tools

Data Ownership and Information Governance (2019-2020)

- Developing the strategy for Content Auto-Classification using ontology and taxonomy for advanced rules-based tagging of content to enable data loss prevention and content findability
- Defining the rules for data loss prevention and records information management through document life-cycle to enable automated system of data governance, with minimal manual input.
- Provides analytics related to content management, re-use, data loss and retention

Key contribution: Was responsible for delivery of the Platform, played a key role in vendor selection, defining requirements, story writing, sprint planning, Semantic data modeling, data review and preparation, front end design and testing. As core team member often interacted with VPs, EDs and Directors of Global Quality Organization to gather requirement and do product demo.
Challenges: Mergers and Acquisition data classification to address data protection, multiple content technologies and integrations, Training SME's on taxonomy build.

Logistics Intelligence platform for On-Demand Supply Chain Data and Analytics (2019)

- Provides on-demand data enabling the perfect order fulfillment.
- Allows near real-time access to KPI's for Logistics Network
- Current Logistics metrics are focused on delivery; Visibility of data will allow optimization of shipper and Transport Service Provider selection
- NLP driven information extraction from structured & unstructured sources
- Groundwork done on the logistics knowledge graph/ontology/taxonomy enables AI and assisted Machine Learning.
- Visualization of logistics routes for faster access and to gain insights
- Provides Joined/integrated data to enable Predictive / Prescriptive analytics

Key contribution: Was responsible for delivery of the Platform, played a key role in vendor selection, defining requirements, story writing, sprint planning, Semantic data modeling, data review and preparation, front end design and testing. As core team member often interacted with VPs, EDs and Directors of Global Quality Organization to gather requirement and do product demo.

Projects at Amgen (continued)

Quality Management System (QMS) Knowledge sharing platform (2018-on-going)

Key contribution: Am currently responsible for delivery of the newly created Quality Management System Reference Data, played a key role in vendor selection, defining requirements, Semantic data modeling, data review and preparation, front end design and testing. As core team member often interacted with VPs, EDs and Directors of Global Quality organization to gather requirement and do product demo.

- Standard taxonomy built to ensure data quality.
- Context based systematic data collection
- Visualization of QMS information for visibility to QMS processes and documentation.

Challenges: Lack of visibility of QMS processes and documentation

Advanced Device Technology (2017)

- Provides a knowledge sharing platform for Advanced Device Technology workers
- Provides search and visualization capability to enable knowledge sharing

Key contribution: Delivered the newly created Device Reference Data, played a key role in vendor selection, defining requirements, Semantic data modeling, data review and preparation, front end design and testing. As core team member often interacted with VPs, EDs and Directors of Global development organization to gather requirement and do product demo.

- Standard taxonomy built to ensure data quality.
- Context based systematic data collection
- Visualization of device information for visibility to data and to gain insights and ideas for device designing

Challenges: All Device Data was in large PDF's not accessible to knowledge workers, not available for analytics

Monsanto Company. (March 15, 1999 – Feb 2016)

Various Roles

Information Intelligence Lead / Ontology Lead	St Louis	Sept 1st, 2012 – Feb 2016
Global Product Lead - Drought and High Oil Corn	St Louis	May 11th, 2009 – August 31st 2012
Sr. Scientist and Data mining and Systems biology group Lead	Bangalore, Karnataka, India	March 15, 1999 to May 9, 2009

Information Intelligence Lead / Ontology Lead

Research Data Integration Platform. Sept 1st, 2012 – Feb 2016

- Outlined the long-term strategy for use of semantic technologies for Agricultural Innovation by developing the plan for the Ag-Intelligence platform by driving consensus through R&D, Global Supply Chain, Corporate Reputation, Global Security.
- Laying out the vision and Launching the Ontology platform- Semaphore for ontology management and utilization and building the infrastructure for Monsanto.
- Planning the foundation to enable innovation in data integration and "smart data mining "by making available an ontology management infrastructure that enables a rich linguistic grounding for areas like qualitative object annotation, text mining, linked data, improved navigation, filtering and visualization of data
- Working with varied groups (Biotechnology, Breeding, Trait integration, Regulatory, Commercial, Public affairs, Prescriptive Farming solutions) to help develop a company standard ontology suited for our seeds and traits business.
- Responsible for creating a suite of Company Standard Ontologies and planning for necessary applications to enable data-interoperability through language for efficient data mining and knowledge generation.
- Responsible for development of workflows to help in creation of a company standard ontology for our seeds and traits business.
- Working as a liaison between Information technologists and biologists to ensure that communication between curators and programmers/ database developers occurs efficiently to facilitate the building and adoption of the company standard ontology.

Global Product Lead Drought and High Oil Corn (May 11th, 2009 – August 31st, 2012)

- Working with varied groups (Biotechnology, Breeding, Trait integration, Regulatory, Commercial, Public affairs) to help launch a drought tolerant product.
- Involved with planning for documentation, dossier preparation, Mode of action with the biotech and the Regulatory manager for drought to facilitate timely handoff to the Regulatory team.
- Involved in project and portfolio planning / forecasting, reporting on sensitivities/risks, NPV and benefits.
- Development of workflows and decision criteria for efficient decision making, documentation and material hand-off at crucial stages of phase change during the product cycle.
- Strategic planning for technology launches globally (Seed Planning and production, Seed Transport, Seed Transportation & Compliance, Site Selection etc.)
- Product Concept Development and valuation; trait quality (Assure products meet quality parameters in terms of alignment to product concept and markets)
- Project Management (e.g. Pipeline management via cross-functional team interactions to meet project timelines).
- Liaison, between external and internal partners, to design, develop, and implement a plan to support new technology introductions into global corn markets.

Work across product teams to provide an integrated product portfolio with abiotic and biotic stress tolerance traits to have a competitive edge.

Sr. Scientist and Data mining and Systems biology group Lead March 15, 1999 to May 9 2009

Genomics Data Mining Platform

Responsible for building the platform for integrating various genomic and phenotypic datatypes for elucidation of Mode of Action.

- Facilitated the integration of varied genomics data in a single discovery platform for pathway building (relationship building), information retrieval and knowledge discovery via ontologies around key genes of interest to the organization and future products.
- Use of multiple types of data to elucidate Mode of Action of genes in corn and soybean and to help in gene selection using systems biology methods

Coordinating / collaborative activities

- Collaboration Manager for Data analysis of data from the Mendel collaboration on transcription factors.
- Collaboration Manager to liaise between Monsanto and the public Plant Ontology Consortium (3 yrs.) towards release of the Monsanto ontology to the public domain for public use (MTA signed, and Plant Anatomy and Trait ontology released).
- Collaborated with Hewlett and Packard to help build the Monsanto's Ontology Server (1 yr.).
- Establishing and maintaining collaboration with IBM and Millennium Pharmaceuticals Inc to evaluate the utility of the IBM - Text Knowledge Mining tool for literature curation and user needs.
- Working as a liaison between Information technologists and biologists to ensure proper implementation of solutions for scientists.

Development of schema for plant genome databases
Provided technical expertise and domain knowledge towards building several plant genome databases @ Monsanto.
Curation of genome literature on Arabidopsis (1223 papers) / Zea mays (40 papers) and Allergens (Allergen:119 sequences).
The job involved being responsible for the quality and quantity of the data content of the different plant genome literature curated databases developed and maintained at Monsanto

Gene selection and prioritization of gene candidates into the pipeline for different trait customers
Nomination of several genes >100 into the product pipeline for the Yield and Stress program

Creating Documentation for efficient transfer of knowledge
(SOP's Regulatory Dossiers, EPA white papers for new technology introduction)

AWARDS

GSK:

- Innovation Pipeline Award – 2020
- Innovation Award Nomination -Logistics Intelligence Platform - 2018

Amgen Inc:

- CIO Award for work on the Reference Data Platform -Amgen – 2017
- Innovation Award Nomination -Logistics Intelligence Platform - 2018

Monsanto Company

- Monsanto Associate Science Fellowship (2006)
- Extra mile award for setting up the SOP for lead gene data integration via GND – Gene Network Database (2009)
- Selected for the Biotech Exchange Program sabbatical (2008)
- Walk the extra mile award (2007) For working on the yield gene GENEX
- Rapid Recognition@ the BSIB function level for the initiative of building a Nomenclature committee across Monsanto (2006).
- Rapid recognition for mobilizing the TOS (The Ontology Server) and (SAR SUPER ANNOTATION REPOSITORY) activities under LEADS from MRC (2005)
- Nominated for the “Above and beyond” award under the category of individual’s contribution to Outstanding Project Implementation in the year 2004 (the nomination was made by colleagues and leaders for conception of the "ontology" based data integration” system, learning the skills, transferring skills and implementing the concept) (2004).
- Rapid recognition for playing host and facilitating better links with individuals from STL (2004)

Patents

- https://www.lens.org/lens/search/patent/list?st=true&q=inventor:(%22AUGUSTINE%20ALICE%20CLARA%22)&p=0&n=10&l=en&preview=true

Publications

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- Alice Clara Augustine & L. D’Souza (1997). Micropropagation of an Endangered Forest Tree - Zanthoxylum rhetsa. Phytomorphology Vol 47(3)
- Alice Clara Augustine & L. D’Souza, (1997) Conservation of Curculigo orchioides an endangered, anticarcinogenic herb. In: Recent Advances in Biotechnological Applications of Plant Tissue and Cell Culture. G. A. Ravishankar & L. V. Venkataraman (eds). Oxford & IBH Publishing Co. Pvt. Ltd. New Delhi, Calcutta 175-179.
- Alice Clara Augustine & L. D’Souza (1997). Somatic embryogenesis in Gnetum ula. National Symposium on Emerging Trends in Plant Tissue Culture and Molecular Biology XX meeting of PTCA (India), Osmania University Jan 29-31
- Alice Clara Augustine & L. D’Souza (2000). Correlation of rate of germination of seeds and callus induction in Gnetum ula. In: Proceedings of the National Symposium held at the Defence Agricultural Research Laboratory, Pithoragarh, UP (India)
- L. D’Souza, Smitha Hegde, Alice Clara Augustine, Rajendra K, Vinitha Cardoza & Kavitha Kulakarni (1997). Cashew is a useful but difficult nut to crack! In: Recent Advances in Biotechnological Applications of Plant Tissue and Cell Culture. G. A. Ravishankar & L. V. Venkataraman (eds). Oxford & IBH Publishing Co. Pvt. Ltd. New Delhi, Calcutta 175-179.
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- Smitha Hegde, L. D’Souza, Icy D’Silva & Alice C. Augustine (1997).Vitro Trees of Cashew (Anacardium occidentale L.)- A Preliminary Evaluation. In: Trends in Plant Tissue Culture and Biotechnology. (Ed) L. K. Pareek (1997) p.239-240.
- L. D’Souza, K Rajendra, Vinitha Cardoza, Alice Clara Augustine, Smitha Hegde & Icy D’ Silva (2003). Micropropagation and conservation of medicinal plants. Role of biotechnology in medicinal and aromatic plants. Ukaaz Publications, Hyderabad.
- L. D’Souza, Alice Clara Augustine, Smitha Hegde, Kavitha Kulakarni, Icy D’Silva (1994). Problems of